

exercise 3.1

This on is fine until you get to the rolling average part. Thankfully there is a library for that. Use pandas to load the data, then use the `rolling().mean()` method to implement. See below.

```
In [4]: import pandas as pd
import matplotlib.pyplot as plt
```

```
In [5]: data3 = pd.read_csv('../data/sunspots.txt', sep='\t', names=['time', 'spots'])
```

```
In [6]: data3
```

```
Out[6]:
```

	time	spots
0	0	58.0
1	1	62.6
2	2	70.0
3	3	55.7
4	4	85.0
...	...	...
3138	3138	16.1
3139	3139	19.6
3140	3140	25.2
3141	3141	23.5
3142	3142	21.6

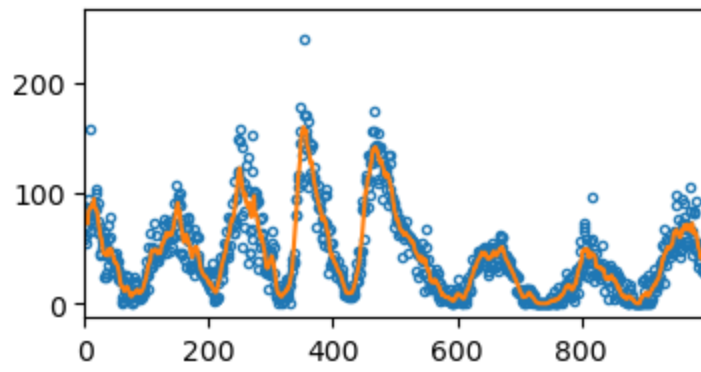
3143 rows × 2 columns

```
In [7]: fig3 = plt.figure(figsize=(4,2))
ax3 = fig3.add_subplot()

ax3.plot(data3['time'], data3['spots'], '.', mfc='none')
ax3.plot(data3['time'], data3['spots'].rolling(10, center=True).mean())

ax3.set_xlim(0,999)
```

```
Out[7]: (0.0, 999.0)
```



In []:

In []:

In []: